

Ligjerata 2b

Nxitimi mesatar dhe i çastit

$$\vec{a}_m = \langle \vec{a} \rangle = \frac{\vec{v}_1 - \vec{v}_0}{\Delta t} = \frac{\Delta \vec{v}}{\Delta t}$$

$$\Delta t \rightarrow 0$$

$$a_m = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \vec{v}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$$

$$\vec{a} = \frac{d}{dt}(v_x \vec{i} + v_y \vec{j} + v_z \vec{k}) = \frac{d}{dt}(v_x \vec{i}) + \frac{d}{dt}(v_y \vec{j}) + \frac{d}{dt}(v_z \vec{k})$$

$$a_x = \frac{dv_x}{dt}; \quad a_y = \frac{dv_y}{dt}; \quad a_z = \frac{dv_z}{dt}$$

komponehet skalore të vektorit $\vec{a}$

$$\vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

